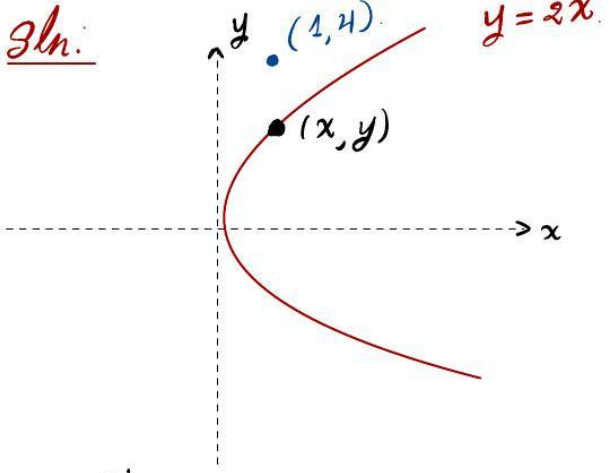
Example. Find the point on the parabola $y^2 = 2x$ that is closest to the point $(1,4)$.

Sln:



Using the formula for the distance between two points $(1,4)$ and (x,y).

Recall: (x_1, y_1) and (x_2, y_2)

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

then, using (x, y) and $(1,4)$.
$x_1\ y_1$ $\quad$ $x_2\ y_2$

$$d = \sqrt{(x-1)^2 + (y-4)^2}$$

Restriction: $y^2 = 2x$. $\quad y = \sqrt{2x}$, $\quad x = \frac{y^2}{2}$.

So,

$$d(x) = \sqrt{(x-1)^2 + (\sqrt{2x} - 4)^2}$$

$$\left(((2x)^{1/2} - 4)^2 \neq \left((2x)^{1/2}\right)^2 - 4^2 \right)$$

$$d(y) = \sqrt{\left(\frac{y^2}{2} - 1\right)^2 + (y-4)^2}$$

$$(x-y)^2 \neq x^2 - y^2$$

Instead of minimizing d, we minimize its square.

$$d^2 = f(y) = \left(\frac{y^2}{2} - 1\right)^2 + (y-4)^2$$

$$f'(y) = 2\left(\frac{y^2}{2} - 1\right)\left(\frac{2y}{2}\right) + 2(y-4)\cdot(1)$$

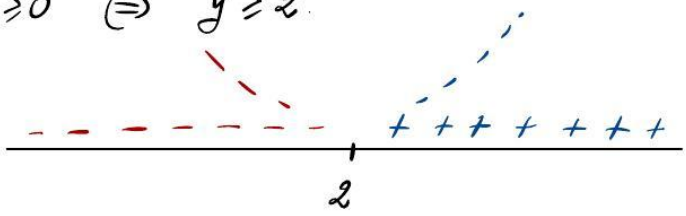
$$f'(y) = (y^2 - 2)(y) + 2y - 8 = y^3 - 2y + 2y - 8 = y^3 - 8.$$

then $f'(y) = 0 \iff y = 2$.

$f'(y) \geq 0 \iff y^3 - 8 \geq 0$

$$(y-2)\underbrace{(y^2 + 2y + 4)}_{\geq 0} \geq 0.$$

then $y-2 \geq 0 \Leftrightarrow y \geq 2$.



By the first derivative test at $y=2$ we have a local minimum. The corresponding value of the x is $x = \frac{1}{2}y^2 = \frac{1}{2}(2)^2 = 2$. then the point on the parabola $y^2 = 2x$ closest to $(1,4)$ is $(2,2)$, the distance between the points is

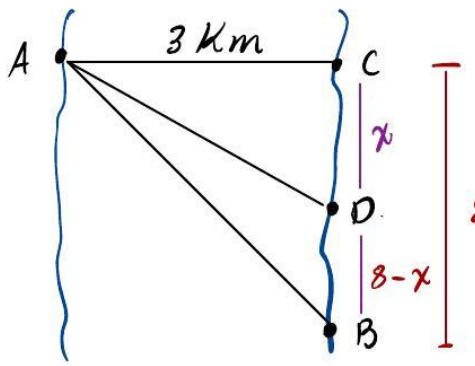
$$d = \sqrt{f(2)} = \sqrt{\left(\frac{2^2}{2} - 1\right)^2 + (2-4)^2}$$

$$= \sqrt{\left(\frac{4}{2} - 1\right)^2 + (-2)^2}$$

$$= \sqrt{(2-1)^2 + 4}$$

$$= \sqrt{5}$$

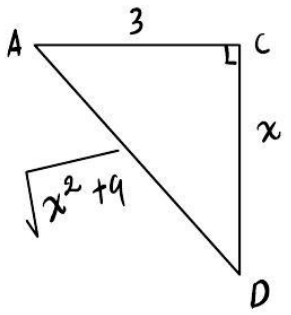
Example. A woman launches her boat from point A on a bank of straight river, 3Km wide, and wants to reach point B, 8 Km downstream on the opposite bank as quickly as possible. She could row her boat directly across the river to point C. and then run to B, or she could row directly to B, or she could row to some point D between C and B and then run to B. If She can row 6 Km/h and run 8 Km/h where should she land to reach B as soon possible.



Sln:

$$\text{time} = \frac{\text{distance}}{\text{rate}}$$

$|BD| = 8-x$, $|AD| = \sqrt{x^2+9}$.



then, the rowing time is $\frac{\sqrt{x^2+9}}{6}$ and the running time is $\frac{(8-x)}{8}$ so, the function for the time is:

$$T(x) = \frac{\sqrt{x^2+9}}{6} + \frac{8-x}{8} \; ; \; \text{Dom } T = [0, 8]$$

Then, $T'(x) = \frac{1}{6}\cdot\frac{d}{dx}\left(\sqrt{x^2+9}\right) + \frac{d}{dx}\left(\frac{8}{8} - \frac{x}{8}\right)$

$$T'(x) = \frac{1}{6}\left\{\frac{1}{2\sqrt{x^2+9}}\cdot\frac{d}{dx}(x^2+9)\right\} + 0 - \frac{d}{dx}\left(\frac{x}{8}\right)$$

$$T'(x) = \frac{1}{6}\left\{\frac{2x}{2\sqrt{x^2+9}}\right\} - \frac{1}{8} = \frac{x}{6\sqrt{x^2+9}} - \frac{1}{8}$$

$T'(x) = 0 \iff \frac{x}{6\sqrt{x^2+9}} - \frac{1}{8} = 0$

$\iff \frac{x}{6\sqrt{x^2+9}} = \frac{1}{8}$

$\iff x = \frac{6}{8}\sqrt{x^2+9}$

$\iff x = \frac{3}{4}\sqrt{x^2+9}$

$\iff x^2 = \left(\frac{3}{4}\sqrt{x^2+9}\right)^2$

$\iff x^2 = \frac{9}{16}(x^2+9)$

$16x^2 = 9x^2 + 81$

$16x^2 - 9x^2 = 81$

$7x^2 = 81$

$x^2 = \frac{81}{7}$

$x = \pm\sqrt{\frac{81}{7}} = \pm\frac{9}{\sqrt{7}}$. then $x = \frac{9}{\sqrt{7}}$

this is a critical point and by the theorem $x=0$ and $x=8$ are also critical points, we follow the closed interval method by evaluating T at all three points

$$T(0) = \frac{3}{6} + 1 = \frac{3}{2} = 1.5$$

$$T\left(\frac{9}{\sqrt{7}}\right) = 1 + \frac{\sqrt{7}}{8} \approx 1.33 \longrightarrow \text{The minimum.}$$

$$T(8) = \frac{\sqrt{3}}{6} \approx 1.42.$$

Since the smallest of these values of T occurs when $x = \frac{9}{\sqrt{7}}$ the absolute minimum value of T must occur there.

Then, the woman should hand the boat at a point $\frac{9}{\sqrt{7}}$ km downstream from her starting point.

Problem for you! Find the area of the largest rectangle that can be inscribed in a semicircle of radius r.

